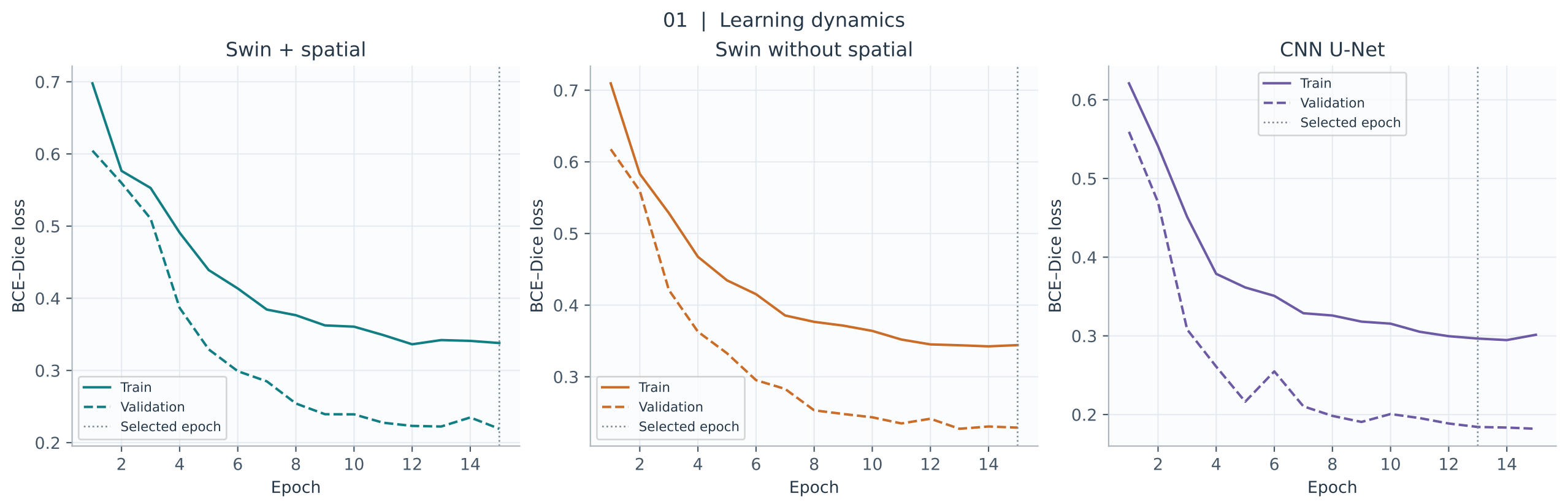
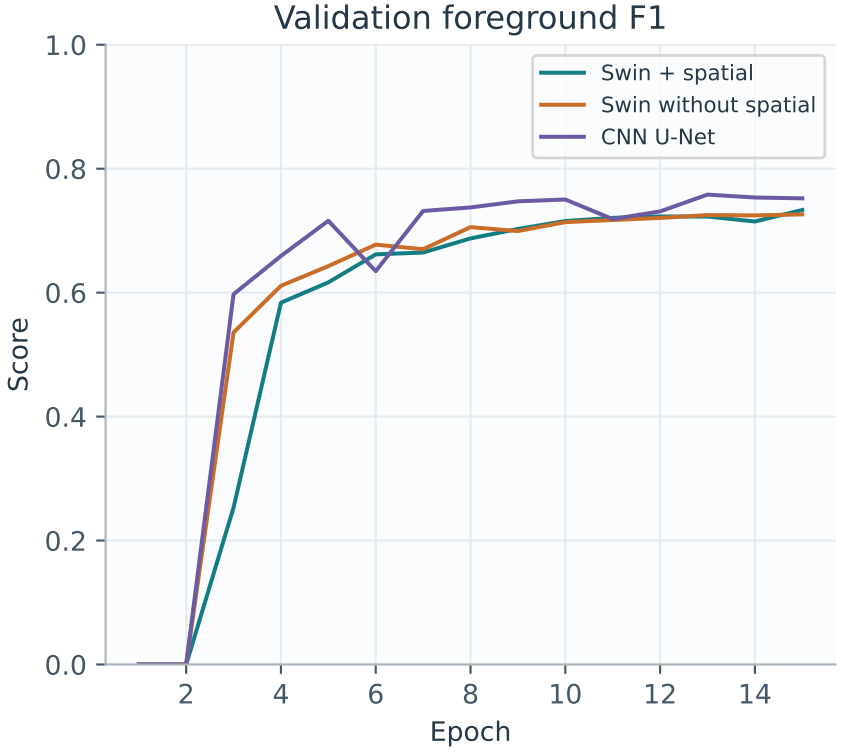
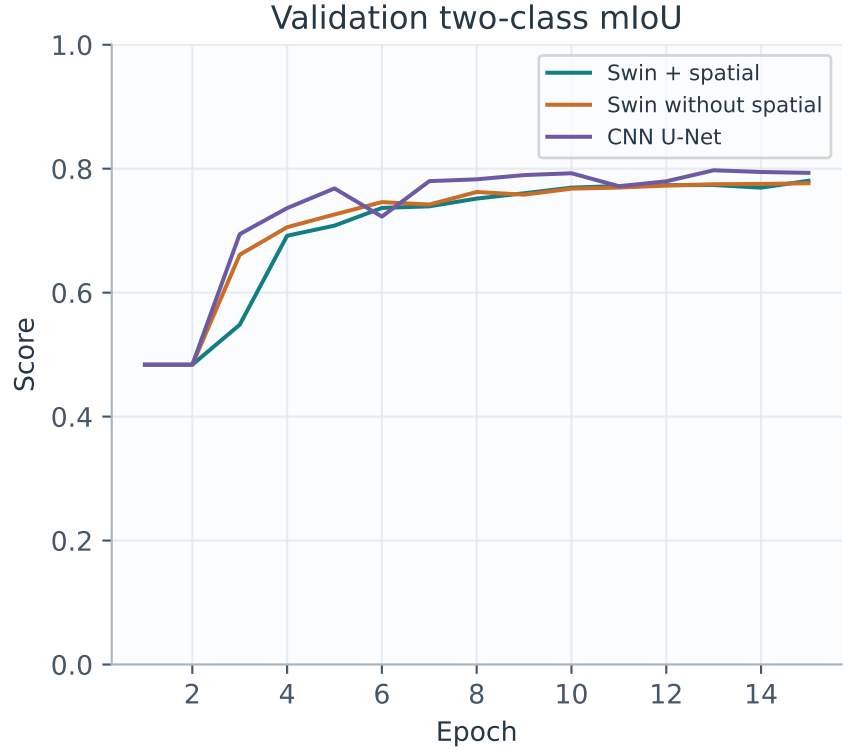
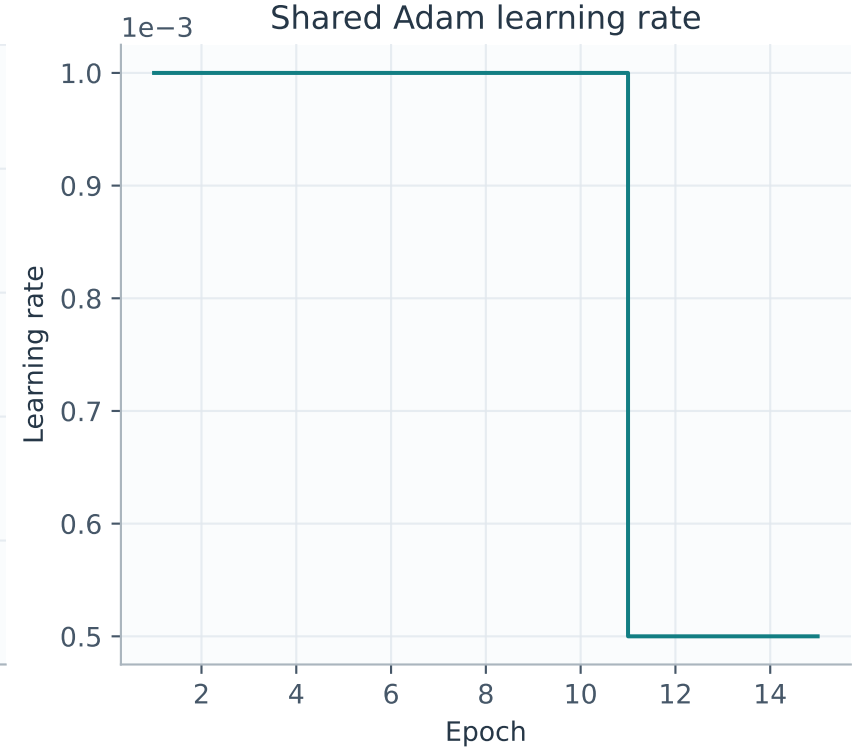
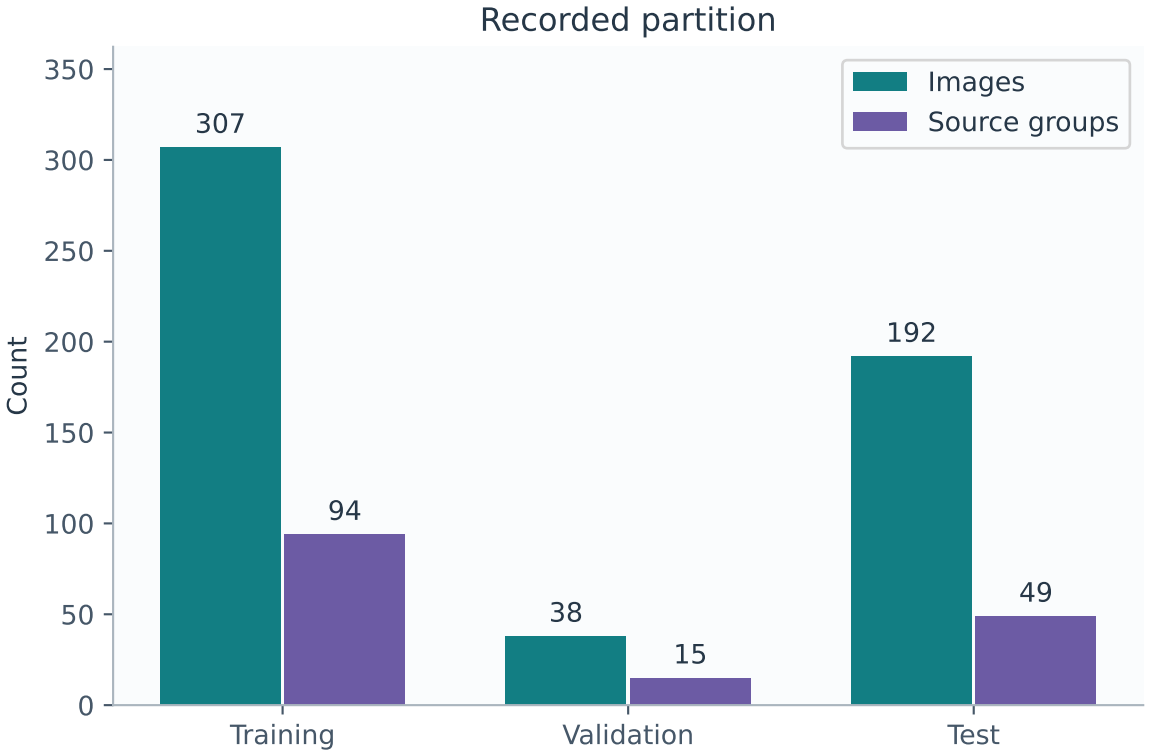
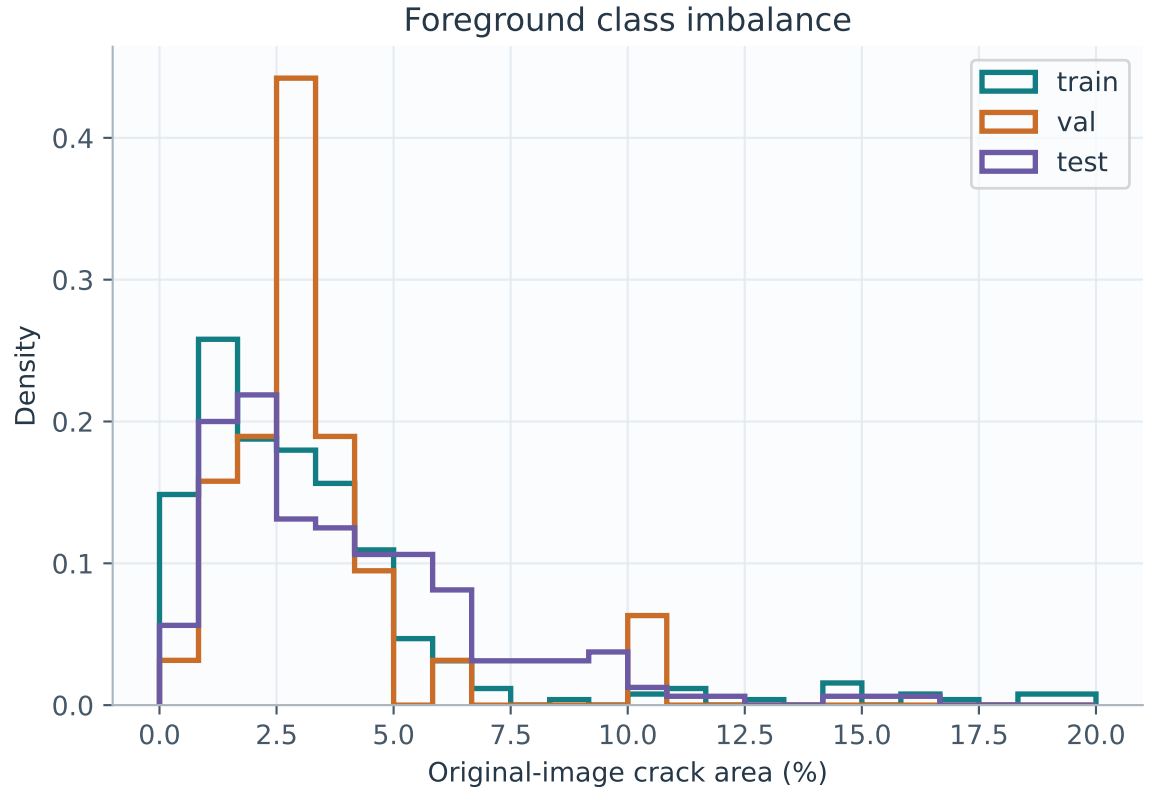

01 | Learning dynamics
Swin + spatial
Swin without spatial
CNN U-Net
BCE–Dice loss
Epoch
Train
Validation
Selected epoch
DeepCrack | exploratory CPU study | measured results | separate from manuscript tables

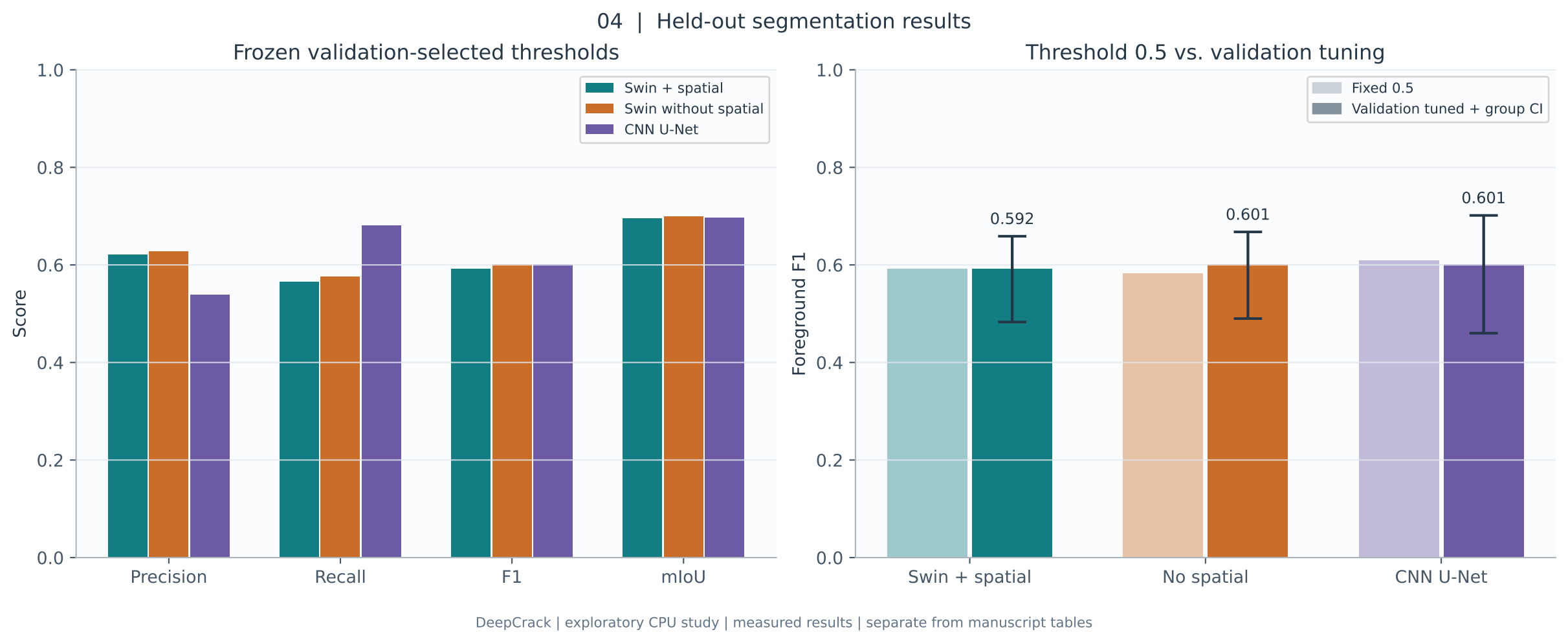

Validation foreground F1
Swin + spatial
Swin without spatial
CNN U-Net
Score
Epoch
1.0
0.8
0.6
0.4
0.2
0.0
2
4
6
8
10
12
14

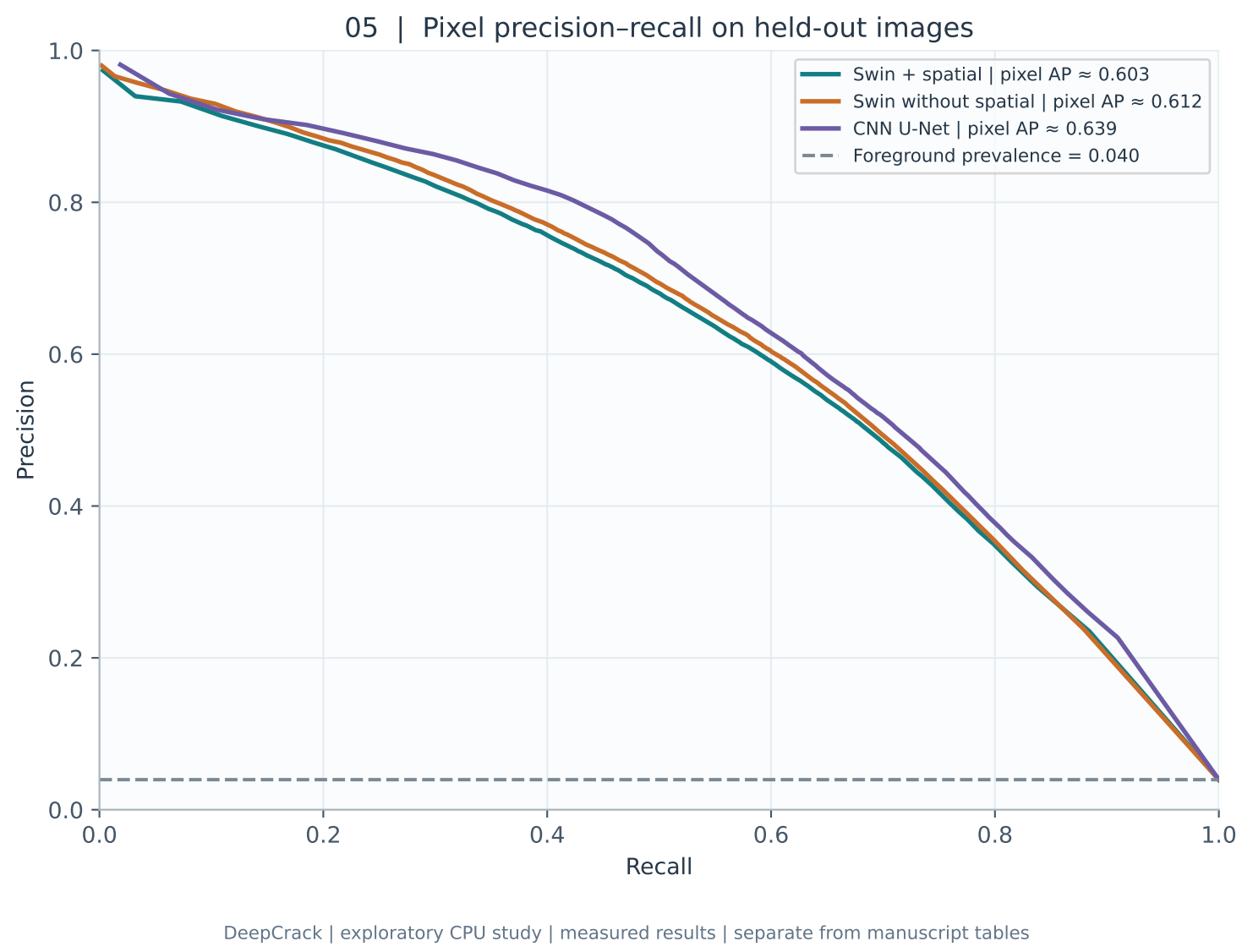

Validation two-class mIoU
Swin + spatial
Swin without spatial
CNN U-Net
Score
Epoch
1.0
0.8
0.6
0.4
0.2
0.0
2
4
6
8
10
12
14

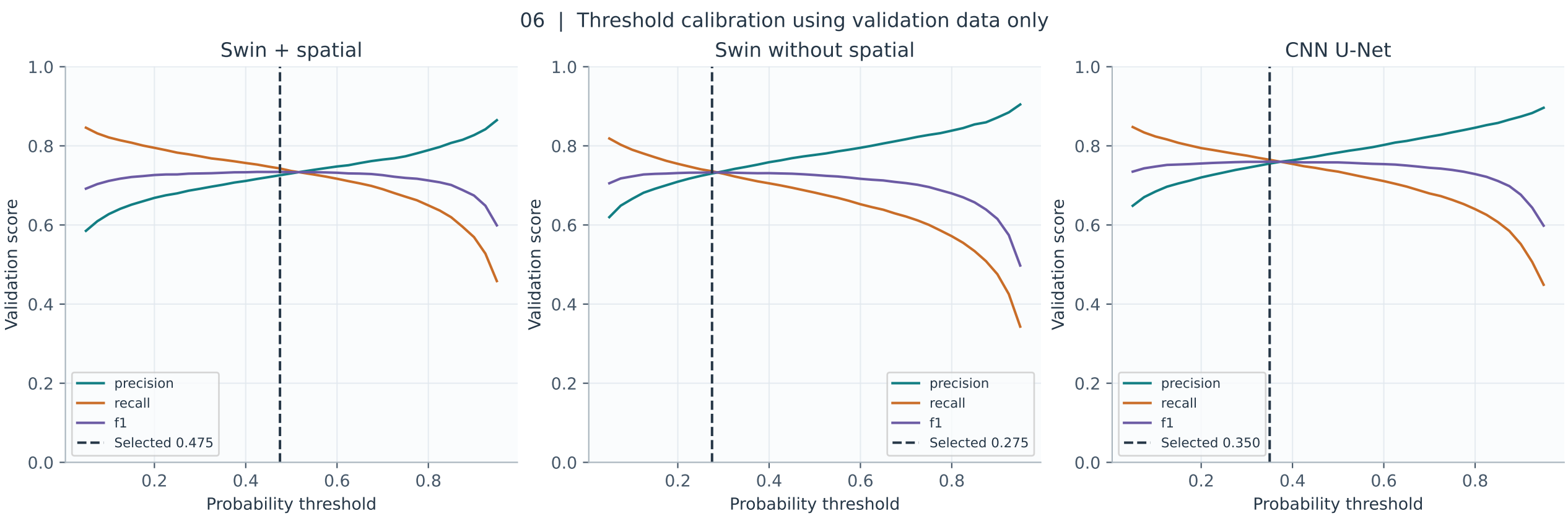

Shared Adam learning rate
1e−3
Learning rate
Epoch
1.0
0.9
0.8
0.7
0.6
0.5
2
4
6
8
10
12
14

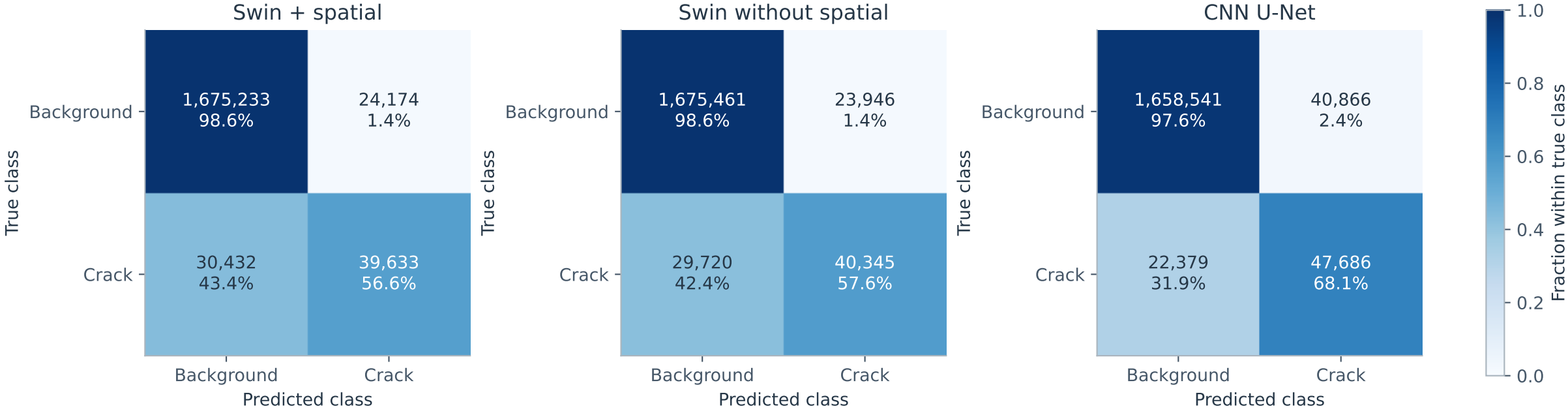

Recorded partition
Images
Source groups
Count
307
94
38
15
192
49
Training
Validation
Test

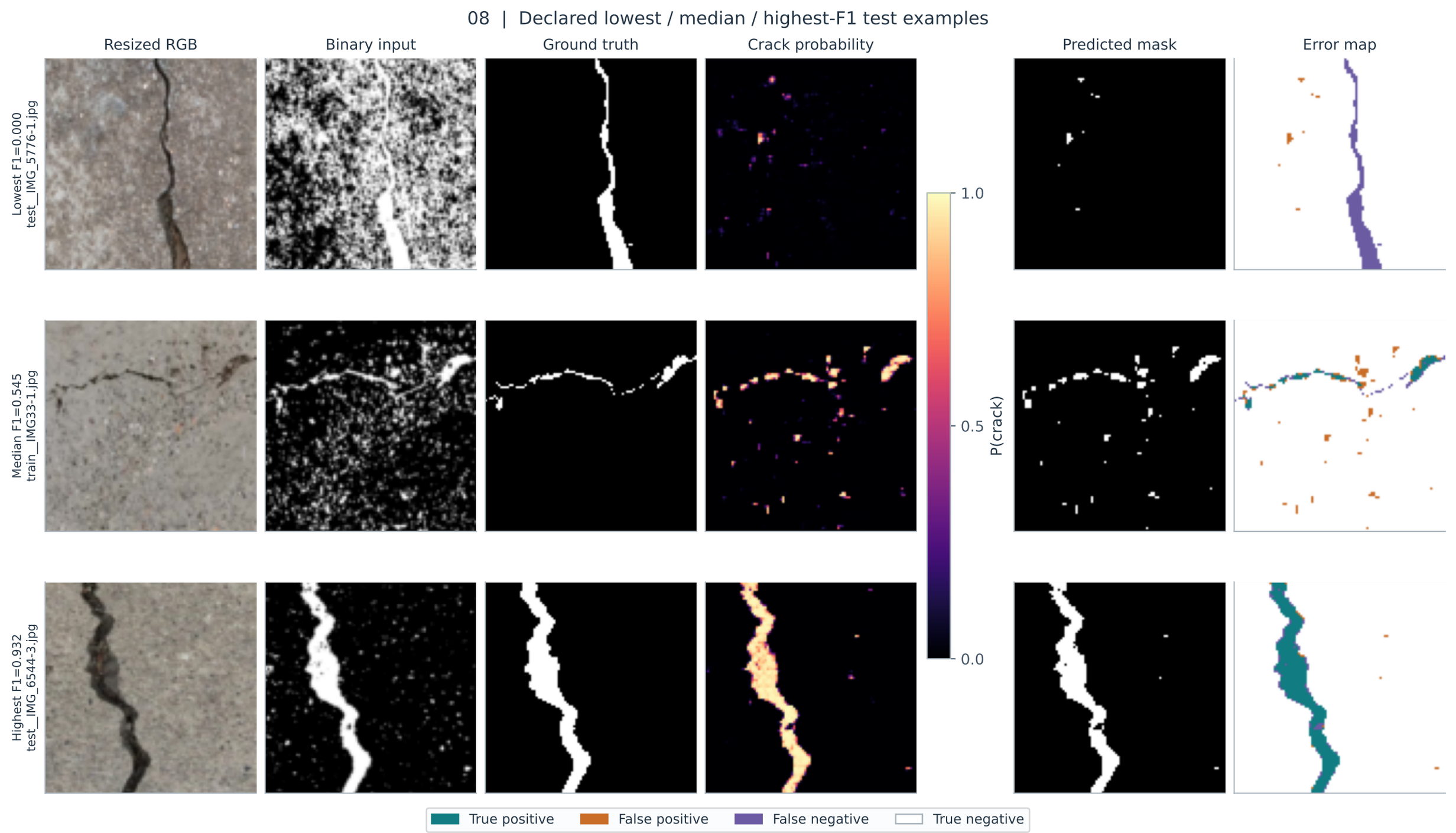

Foreground class imbalance
train
val
test
Density
Original-image crack area (%)

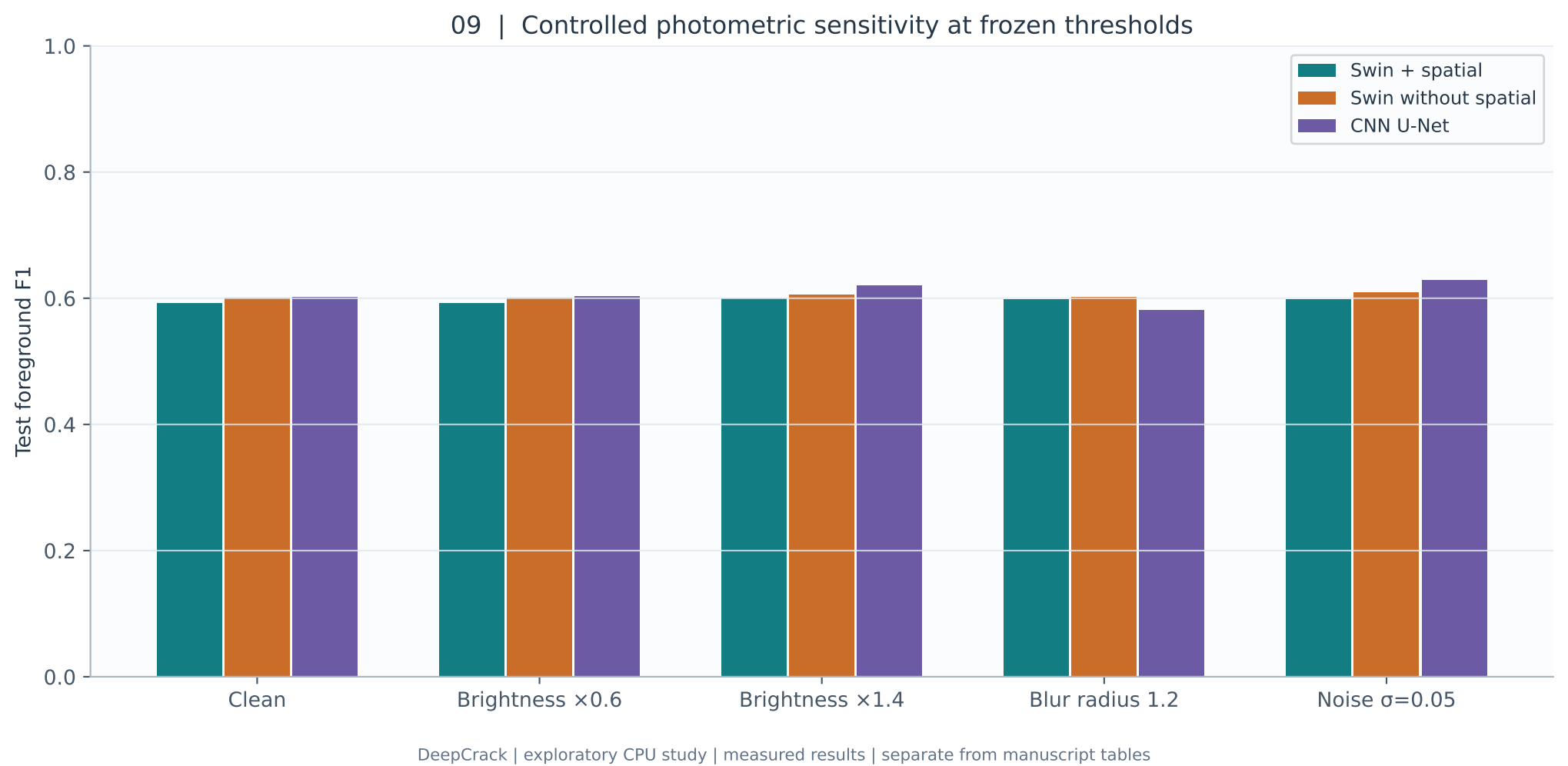

04 | Held-out segmentation results
Frozen validation-selected thresholds
Swin + spatial
Swin without spatial
CNN U-Net
Score
1.0
0.8
0.6
0.4
0.2
0.0
Precision
Recall
F1
mIoU
Threshold 0.5 vs. validation tuning
Fixed 0.5
Validation tuned + group CI
Foreground F1
0.592
0.601
0.601
Swin + spatial
No spatial
CNN U-Net
DeepCrack | exploratory CPU study | measured results | separate from manuscript tables

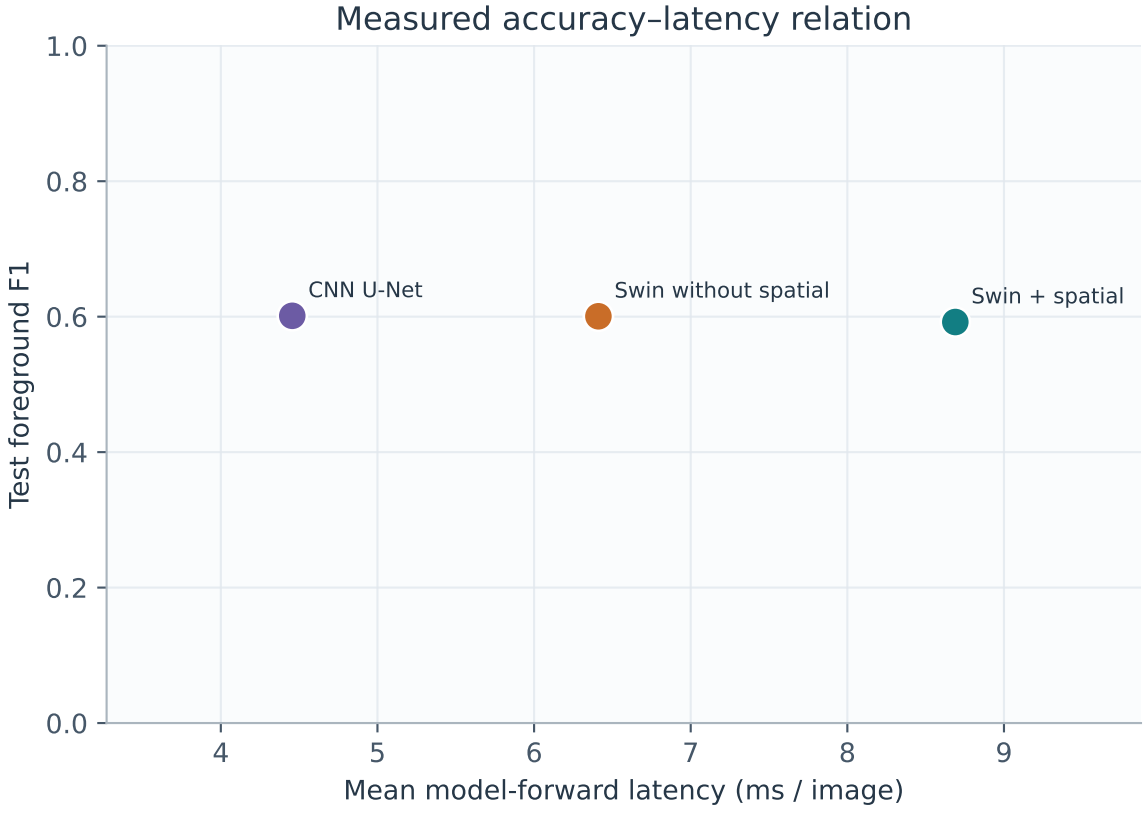

05 | Pixel precision–recall on held-out images
Swin + spatial | pixel AP ≈ 0.603
Swin without spatial | pixel AP ≈ 0.612
CNN U-Net | pixel AP ≈ 0.639
Foreground prevalence = 0.040
Precision
Recall
1.0
0.8
0.6
0.4
0.2
0.0
0.0
0.2
0.4
0.6
0.8
1.0
DeepCrack | exploratory CPU study | measured results | separate from manuscript tables

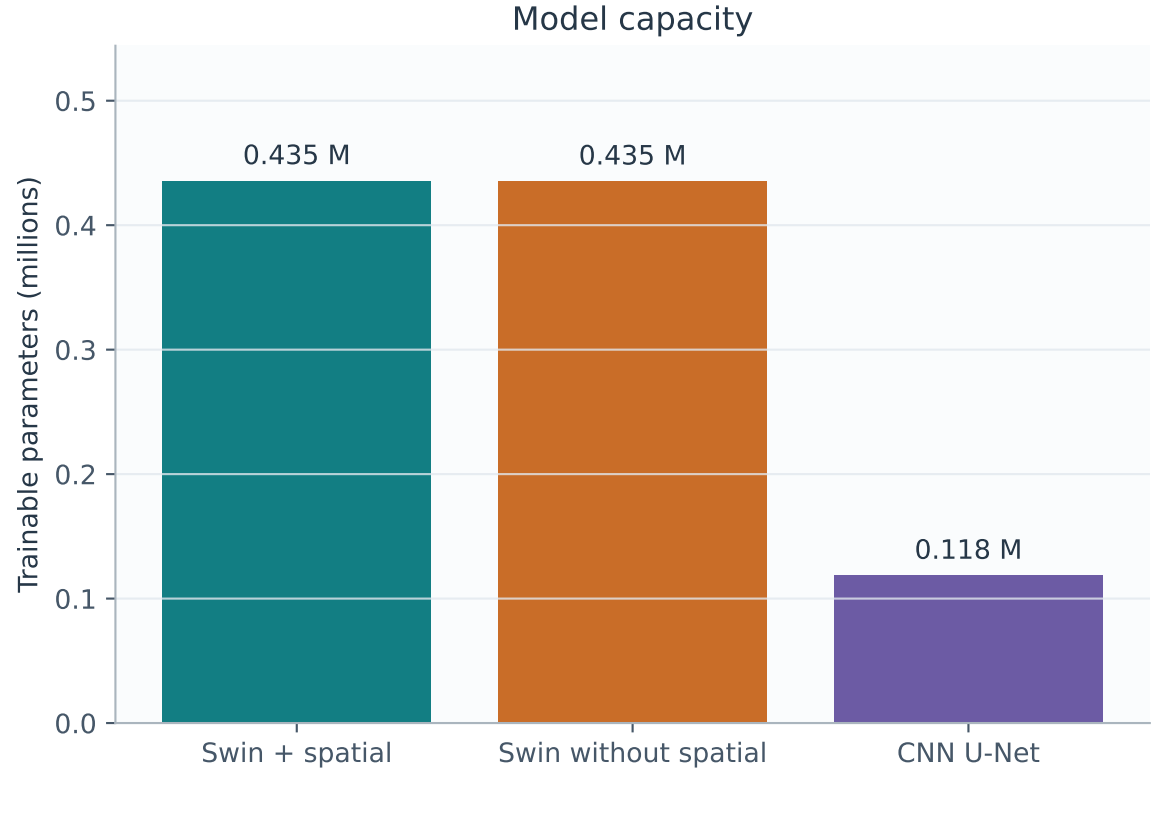

06 | Threshold calibration using validation data only
Swin + spatial
Swin without spatial
CNN U-Net
Validation score
Probability threshold
precision
recall
f1
Selected 0.475
Selected 0.275
Selected 0.350
0.0
0.2
0.4
0.6
0.8
1.0


Swin + spatial
Swin without spatial
CNN U-Net
True class
Predicted class
Background
Crack
1,675,233
98.6%
24,174
1.4%
30,432
43.4%
39,633
56.6%
1,675,461
98.6%
23,946
1.4%
29,720
42.4%
40,345
57.6%
1,658,541
97.6%
40,866
2.4%
22,379
31.9%
47,686
68.1%
Fraction within true class
1.0
0.8
0.6
0.4
0.2
0.0


08 | Declared lowest / median / highest-F1 test examples
Resized RGB
Binary input
Ground truth
Crack probability
Predicted mask
Error map
Lowest F1=0.000
test__IMG_5776-1.jpg
Median F1=0.545
train__IMG33-1.jpg
Highest F1=0.932
test__IMG_6544-3.jpg
1.0
0.5
0.0
P(crack)
True positive
False positive
False negative
True negative
DeepCrack | exploratory CPU study | measured results | separate from manuscript tables


09 | Controlled photometric sensitivity at frozen thresholds
Swin + spatial
Swin without spatial
CNN U-Net
Test foreground F1
1.0
0.8
0.6
0.4
0.2
0.0
Clean
Brightness ×0.6
Brightness ×1.4
Blur radius 1.2
Noise σ=0.05
DeepCrack | exploratory CPU study | measured results | separate from manuscript tables


Measured accuracy–latency relation
Test foreground F1
1.0
0.8
0.6
0.4
0.2
0.0
CNN U-Net
Swin without spatial
Swin + spatial
4
5
6
7
8
9
Mean model-forward latency (ms / image)


Model capacity
Trainable parameters (millions)
0.5
0.4
0.3
0.2
0.1
0.0
0.435 M
0.435 M
0.118 M
Swin + spatial
Swin without spatial
CNN U-Net